COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE009	**SECTION:**	CPE21S1
DATE PERFORMED:	7/25/2026	**DATE SUBMITTED:**	7/25/2026
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ACTIVITY NO. _4.1_ [Array-Based vs. STL Stack Implementations]

1. PROCEDURE OUTPUT

HOA421 > main.cpp > ...

```cpp
#include <iostream>
#include <stack> // calling stack from the STL

int main() {
    // create an object for the stack
    std::stack<int> newStack;

    std::cout << "Stack Empty? " << newStack.empty() << std::endl;
    std::cout << "The size of the stack: " << newStack.size() << std::endl;

    newStack.push(3);
    newStack.push(8);
    newStack.push(15);

    std::cout << "The top of the element is: " << newStack.top() << std::endl;
    std::cout << "Removing elements from the stack is: \n";
    newStack.pop();
    std::cout << "The top of the element is: " << newStack.top() << std::endl;
    newStack.pop();
    std::cout << "The top of the element is: " << newStack.top() << std::endl;
    newStack.pop();
    std::cout << "Stack Empty? " << newStack.empty() << std::endl;

    return 0;
} <- #4-25 int main()
```

```
Stack Empty? 1
The size of the stack: 0
The top of the element is: 15
Removing elements from the stack is:
The top of the element is: 8
The top of the element is: 3
Stack Empty? 1
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

CODE

STACKS > main.cpp > ...

```cpp
#include <iostream>

const size_t maxCap = 100;
int stack[maxCap]; // stack with max of 100 elements
int top = -1, i, newData;

void push();
void pop();
void Top();
bool isEmpty();
bool isFull();
void display();


int main() {
    int choice;
    std::cout << "Enter number of max elements for new stack (max 100): ";
    std::cin >> i;

    while (true) {
        std::cout << "\nStack Operations: " << std::endl;
        std::cout << "1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY" << std::endl;
        std::cin >> choice;

        switch (choice) {
            case 1:
                push();
                break;
            case 2:
                pop();
                break;
            case 3:
                Top();
                break;
            case 4:
                std::cout << (isEmpty() ? "1 (True)" : "0 (False)") << std::endl;
                break;
            case 5:
                std::cout << (isFull() ? "1 (True)" : "0 (False)") << std::endl;
                break;
            case 6:
                display();
                break;
            default:
                std::cout << "Invalid Choice." << std::endl;
                break;
        }
    }
    return 0;
}
```

```
52  bool isEmpty() {
53      return top == -1;
54  }
55
56  bool isFull() {
57      return top == i - 1;
58  }
59
60  void push() {
61      // check if full -> if yes, return error
62      if (isFull()) {
63          std::cout << "Stack Overflow." << std::endl;
64          return;
65      }
66      std::cout << "New Value: ";
67      std::cin >> newData;
68      stack[++top] = newData;
69  }
70
71  void pop() {
72      // check if empty -> if yes, return error
73      if (isEmpty()) {
74          std::cout << "Stack Underflow." << std::endl;
75          return;
76      }
77      // display the top value
78      std::cout << "Popping: " << stack[top] << std::endl;
79      // decrement top value from stack
80      top--;
81  }
82
```

```cpp
void Top() {
    if (isEmpty()) {
        std::cout << "Stack is Empty." << std::endl;
        return;
    }
    std::cout << "The element on the top of the stack is " << stack[top] << std::endl;
}

void display() {
    if (isEmpty()) {
        std::cout << "Stack is Empty." << std::endl;
        return;
    }
    std::cout << "Stack (top to bottom): ";
    for (int j = top; j >= 0; j--) {
        std::cout << stack[j] << " ";
    }
    std::cout << std::endl;
}
```

Output

```
Enter number of max elements for new stack (max 100): 4

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
1
New Value: 3

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
1
New Value: 1

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
1
New Value: 5

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
1
New Value: 7

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
1
Stack Overflow.

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
5
1 (True)
```

```
Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
4
0 (False)

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
3
The element on the top of the stack is 7

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
6
Stack (top to bottom): 7 5 1 3

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
2
Popping: 7

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
6
Stack (top to bottom): 5 1 3

Stack Operations:
1. PUSH, 2. POP, 3. TOP, 4. isEMPTY, 5. isFULL, 6. DISPLAY
```

conclusion
The process of coding this program gave me a deeper insight into the actual mechanism of the stack using only an array and an index. The most important thing for me was the necessity to include an edge case in virtually all functions; for example, in case of push operation it is necessary to verify if the stack is full and in pop – if the stack is empty; otherwise, one would be accessing some elements which do not actually contain any valid values. What is more, I found it rather convenient to have isEmpty and isFull functions as pure boolean functions which were used everywhere in the code.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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